

DiaTool-DPO: Multi-Turn DPO for Tool-Augmented LLMs

Enhancing Tool-Augmented LLM Dialogue via
Direct Preference Optimization

Automatically constructing paired
trajectory datasets to improve slot-
filling and tool call rejection without

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

- TA-LLMs trained only with SFT struggle with incomplete queries and out-of-scope requests.
- This leads to hallucinated tool calls and poor dialogue control.
- Three key failure modes of SFT-only models:
 - • Failing to ask follow-up questions for missing arguments.
 - • Making hallucinated tool invocations.
 - • Failing to reject inappropriate or unavailable tool calls.
- Motivation: Preference optimization provides a robust solution to enforce correct conversational flows without relying purely on SFT trajectories.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

Interactions are modeled as an MDP with 5 unobserved internal states:

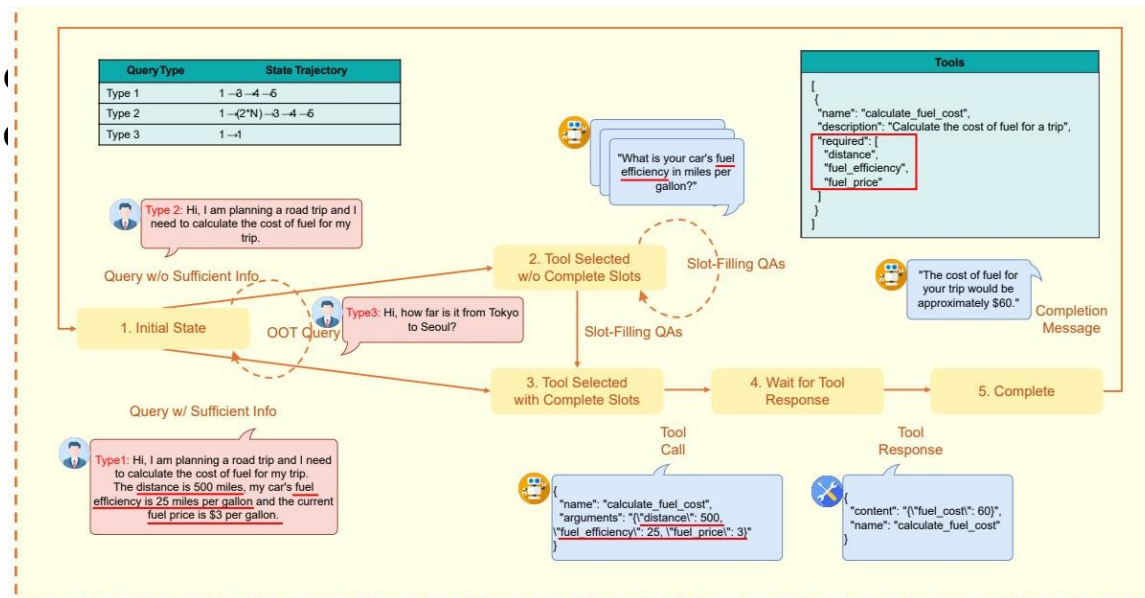
Initial State: No history. Out-of-scope requests return rejection here.

Tool Selected w/o Complete Slots: Missing parameters. Slot-filling QA

Tool Selected w/ Complete Slots: Tool and all arguments are determined

Wait for Tool Response: Tool executed, awaiting results.

Complete: Tool results received, completion message generated.



Three Query Types Define Distinct State Trajectories

User queries are classified into three types based on their state transition trajectories:

Query Type	State Trajectory	Description & Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query contains all required arguments. No slot-filling needed.
Type 2	$1 \rightarrow (2 * N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query lacks some arguments. Slot-filling QA repeats N times until all fields are gathered.
Type 3	$1 \rightarrow 1$	Requested functionality is unavailable. Tool call must be rejected.

Automatic Paired Trajectory Construction Without Human Labels

Labels

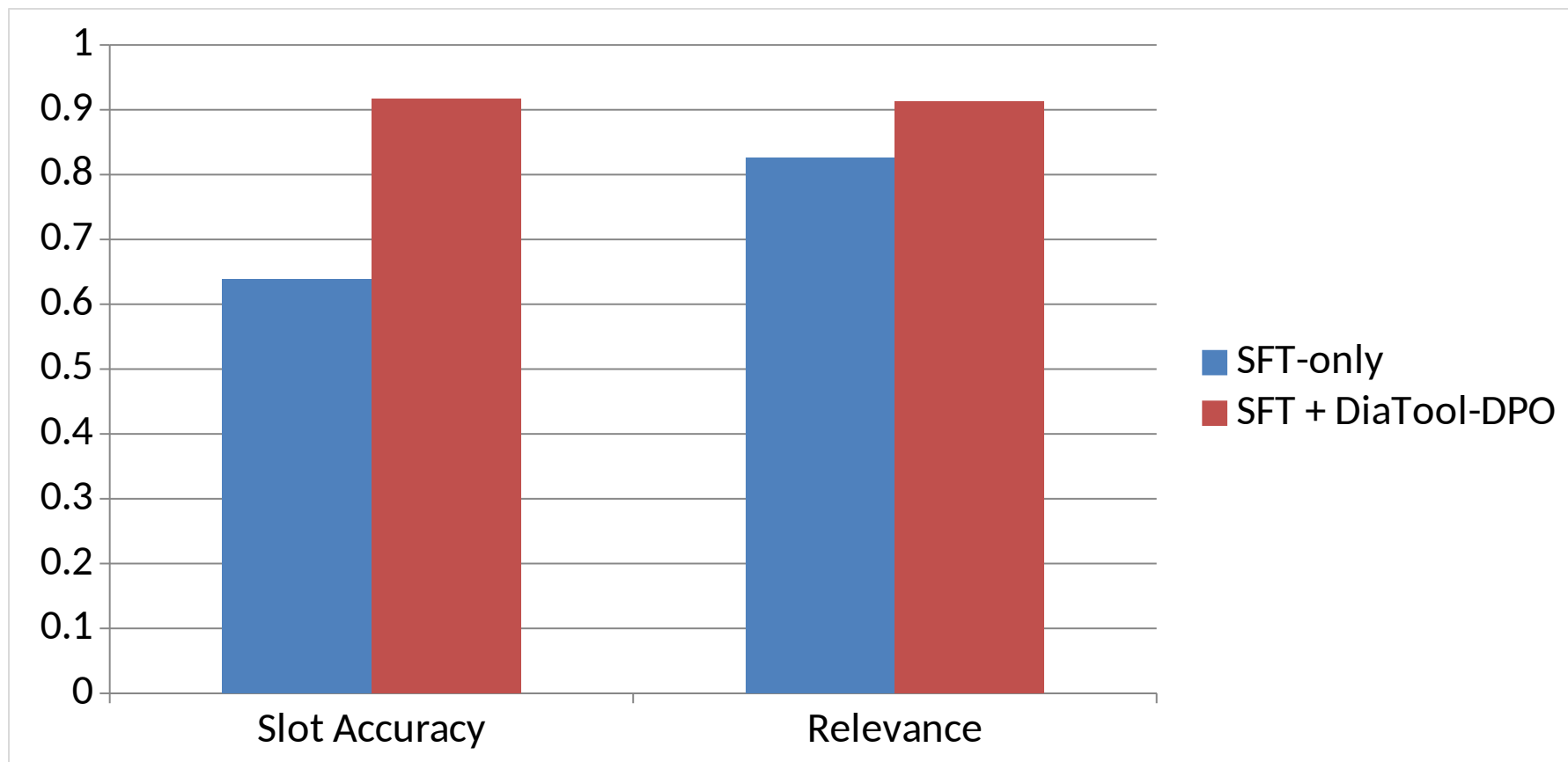
The dataset automatically pairs chosen (correct) and rejected (incorrect) trajectories to teach dialogue control lessons:

Query	Chosen traj.	Rejected traj.	Learning lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089/562	Easy, Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg.	Macro Avg.
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

- MDP formulation enables systematic trajectory pairing without human annotation.
- Automatic dataset construction eliminates the need for expensive human labeling.
- DPO strongly complements SFT for dialogue control (DPO alone is insufficient).
- The approach is language-agnostic, demonstrating broad applicability for production.
- Opens the path to achieving GPT-4o-level conversational tool use on smaller open-source models.